

Representation Convergence · Complete Experimental Summary

Testing the Platonic Representation Hypothesis: do independently trained encoders converge to one structure? All numbers verified against the final report. 34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful.

1 · The encoders · architecture, how each was pretrained, and its role

encoder / dim	architecture	objective + supervision	role and measured result
DINOv2 s/b/l 768/1536/2048	ViT	Self-distillation (DINO+iBOT); SELF-SUPERVISED	Hub members + capacity ladder: 40.9/53.5/58.3% of ceiling; +15.3 pts/decade, R ² 0.949. img_small sets hub width.
bge-m3 1024	XLM-R	Contrastive on mined pairs; PAIR-SUPERVISED (text only)	Hub member + head TARGET. Top pair with SBERT (p 0.771).
GPT-2 768	Transformer (causal)	Next-token prediction; SELF-SUPERVISED	The collapse case: eff.rank 6.6, readable 95.1% / writable 24.8%. Whitening: R@1 0.130→0.478 = 102.6% of contrastive (C.11).
BERT 768	Transformer (masked)	Masked LM; SELF-SUPERVISED	Hub member, ladder rung. eff.rank 65.0.
SBERT 768	BERT + NLI	NLI entailment pairs; SUPERVISED	Best head target: 101.0% of native. Healthiest text space (pair-cos 0.135).
SigLIP 2 768	ViT	Image-text sigmoid; PAIR-SUPERVISED (HELD OUT)	Cross-lineage transfer 94.2%, inside 93.8–96.5 band.
ConvNeXt-base 1024	CONVNET	ImageNet-22k classes; FULLY SUPERVISED (HELD OUT)	96.7% of native — ABOVE the band. Architecture does not bound the claim.
MobileCLIP-S1 / SigLIP 2 (Exp B)	ViT (both)	Contrastive; PAIR-SUPERVISED	Experiment B pair: phone→server adapter. cos 0.900, R@1 0.987, AUC 0.999.
DINOv2-giant 3072 (cls+patch)	ViT	Self-distillation; SELF-SUPERVISED	Scale-extension top rung. p 0.339 < large's 0.418. P2 FALSIFIED. Cannot be hub member ($\gamma=1.26$).
Qwen3-Emb 0.6B/4B	Qwen3 (causal, FT)	Causal LM FT for embed; CONTRASTIVE FT	Scale-extension text ladder. p 0.352 / 0.307. P3 FALSIFIED.
FractalDB ResNet-50 2048	ConvNet	FractalDB-1k only; no natural images	Fractal probe (G9): R = 0.09 (raw), 0.11 (whitened). G10 transfer: 0.014 R@1 — at floor, = random twin.
FractalDB DeiT-tiny 384	ViT	FractalDB-1k only; no natural images	Fractal probe (G9): R = 0.21. G10 transfer: 0.020 R@1 — at floor, = random twin.

Source: Report §E.10 (provenance), §4/C.11/C.13/E.16 (results), Appendix F (fractal probe).

2 · The experiments · what each tested, how, and what it found

experiment	question	method	result
A WikiText	Do two text encoders share structure on passages?	10,000 WikiText passages; fit one linear map	R ² mean 0.737, peak 0.797. Robust to caption protocol.
B MobileCLIP→SigLIP	Can a phone encoder map into a server space?	4,000 COCO; ridge + Procrustes; Core ML export	cos 0.900, R@1 0.987, AUC 0.999. 88.4% of ceiling. MLP = ridge (0.9183).
C / vec2vec	Recover correspondence from shape alone?	Gromov-Wasserstein + CSLS self-learning	39.7% exact (595× chance). Whitening DESTROYS it (→0.1%).
E-series ladder	Does raw agreement order by isotropy?	7 encoders pairwise; eff.rank, whitening rescue	GPT-2 collapsed → whitening 102.6%. Isotropy is what similarity training supplies.
G-series hub	Can ONE frozen space serve 7 encoders?	512-d whitened PCA of 7 concat. spaces; one ridge map per encoder	42/42 pairs at 103.6%. Head trained on ONE encoder reads others at 93–96%.
G4 held-out	Does DINOv2 ancestry explain the transfer?	SigLIP + ConvNeXt (no lineage) into FROZEN hub	SigLIP 94.2%, ConvNeXt 96.7% — ancestry not doing the work.
G11–G16 width cliff	Why does transfer collapse past hub width 768?	Sweep hub width; test 8 accounts; pre-register locations	SEVEN accounts falsified, eighth (RANK WALL) confirmed to single-direction.
Scale ext. E.16	Does alignment strengthen with capacity?	DINOv2-giant + Qwen 0.6B/4B; 11 encoders, pre-registered	P2 & P3 FALSIFIED: both ladders rise then FALL. Not monotonic.
G8 similarity	Do four map-free metrics agree on the roster?	mutual k-NN, debiased CKA, relative rep., shape-p; raw+whitened	Cross-modal raw ρ 0.288; whitened 0.062. bge–SBERT 0.771. GPT-2 outlier.
G9 fractal probe	Does fractal-only training create shared geometry?	FractalDB ResNet-50 + DeiT vs 8 encoders; brackets + R	R ≈ 0.09 (CNN) / 0.21 (ViT). P1/P2 CONFIRMED. P3 INCONCLUSIVE. P5-cnn FALSIFIED (raw), P5w CONFIRMED.
G10 hub transfer	Is fractal geometry recoverable through the hub?	Fractal → entry map → frozen hub → frozen head → bge R@1	0.014/0.020 R@1, AT FLOOR. H1 FALSIFIED. No training-specific gain beyond random init.

Every experiment pre-registered its criterion where a verdict was possible. Negatives reported with the same weight as positives.

3 · The measurements, and the headline numbers

measurement	what it asks	chance / reference
raw cross-space cosine	Same DIRECTION?	≈ 0 (high-dim random vectors orthogonal)
Spearman ρ of pairwise dist.	Same ordering of items? (no map)	the PRH alignment measure
kNN overlap (k=10)	Same nearest neighbours?	chance 0.010; measured 0.62 = 62×
R@1 / R@5 (retrieval)	Correct item ranks first / top 5?	chance 0.001 = 1/1000 gallery
% of native head	How much of a native head does a frozen head recover?	100% = frozen head loses nothing
verification AUC	Can one threshold separate true from random pairs?	0.5 = chance
effective rank	How many directions carry variance?	full = healthy; GPT-2 = 6.6
mean pair-cosine	Is one direction crowding the space?	≈ 0 healthy; 0.999 = collapsed
reality fraction R	Where on the random→ImageNet scale does fractal land?	0 = random; 1 = ImageNet twin

Headline results

Convergence is REAL but LOCAL. Encoders agree on neighbourhoods (kNN 62× chance, ρ up to 0.77) while raw coordinates are orthogonal ($\cos \approx 0$). Different charts of one manifold, not one space.

The strongest evidence is a FAILURE. Seven explanations of the width cliff were pre-registered and falsified; the eighth (rank wall) confirmed. A test that could have killed the account, and didn't.

Objective > lineage > modality. bge-SBERT (0.77, shared OBJECTIVE, no lineage) beats img-img (0.56, shared ViT ancestry). A shared training goal aligns more than shared architecture.

Capacity does NOT monotonically help. Both ladders rise then fall: image 0.418→0.339 at giant, text 0.352→0.307 at Qwen-4B. Pre-registered, both FALSIFIED.

Isotropy is a FRAME, not content. Whitening rescues retrieval (0.130→0.478) but destroys unsupervised matching (39.7%→0.1%). Same transform, opposite effect.

Natural-world data is load-bearing. Fractal training creates real shared geometry ($R \approx 0.09$ –0.21, P2 confirmed) but that geometry does NOT transfer through the frozen hub (R@1 at floor, = random twin, H1 falsified). Visible geometry ≠ recoverable structure.

Project verdict ledger — 34 pre-registered prediction units

verdict	count	includes
CONFIRMED	21	learned geometry, cross-modal/lineage transfer, hub 93–96%, fractal geometry above random (P1/P2)
FALSIFIED	7	fractal geometry recoverable through hub (it was not); naive scaling beyond 304M; MLP > ridge
INCONCLUSIVE	4	whether natural-world dependence differs between image and text encoders (P3 CIs straddle zero)
NOT MEANINGFUL	2	H3 CNN/ViT %-of-native ratios blocked by the 0.050 floor guard; both transfer and native at floor

4 · The cat / dog intuition · one worked example

The setup. Imagine a tiny 2-D embedding of animal photos: cats cluster on one side, dogs on the other, and the whole cloud is stretched — wide in one direction, thin in the perpendicular one. This toy stands in for what every encoder in the project produces: a point cloud whose shape carries the meaning.

THE EXPERIMENT — PCA then whitening

PCA finds the cloud's directions of maximum variance (the eigenvectors) and how much variance each carries (the eigenvalues). Here the horizontal eigenvalue is large, the vertical one small. Whitening divides each direction by the square root of its eigenvalue, so every direction has variance 1 — the stretched cloud becomes round.

THE MEASUREMENT — what changes, what doesn't

Cosine similarity reads the ANGLE between vectors. In the stretched cloud, the wide direction dominates every angle, so cosine mostly measures that one axis. After whitening, all directions count equally, so cosine reads the full shape. We measure this by retrieval R@1 and by mean pair-cosine before and after.

THE RESULT — the frame was the problem

The cat/dog SEPARATION was in the cloud all along. Whitening did not create it — it removed the stretch that was hiding it. This is exactly the GPT-2 result (C.11): a collapsed text space, unreadable by cosine, is rescued to 102.6% of a contrastively trained encoder by whitening alone. No new information, just a corrected coordinate frame.

THE CAVEAT — whitening is not free

Whitening does NOT make cats and dogs identical — it changes the coordinates, never which items are distinct; the two clusters stay separate. And it CUTS BOTH WAYS: the uneven variance it removes is noise when READING a known correspondence (retrieval improves) but SIGNAL when FINDING an unknown one (unsupervised matching collapses, C.12). Isotropy helps one task and destroys the other.

THE FRACTAL EXTENSION — same logic, different lesson

The fractal probe (G9/G10) adds one more turn to this logic. The fractal CNN's embedding is stretched like the cat/dog cloud (mean pair-cosine 0.85). Whitening exposes a small real signal (P5w confirmed, R_{white} ≈ 11%) — the frame WAS hiding something, just as with GPT-2. But when that geometry is pushed through the hub's fitted entry map and frozen head, it transfers at the floor (R@1 0.014 = random twin). The frame was hiding real geometry, but that geometry was not enough for semantic function. Visible ≠ recoverable: the strongest lesson of the fractal thread, and a new implication for the field.

Why this example anchors the whole project. Every headline follows the same logic as cats and dogs: the shared structure is present in the raw spaces (shape), but the coordinate FRAME decides whether a simple metric can see it. Convergence is real and local; the frame — not the content — is what training supplies, and equalising it is a trade, not a universal fix. The fractal probe adds: and natural-world content is what makes the structure useful.

Full step-by-step: [PCA_Cat_Dog_All_Graph_States.pdf](#).